

Test Report - VxMark Continuous Operations Stress Test

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Summary

The VxMark system passed stress testing in 168 hours of continuous operation (across 8 days) in cold and warm conditions internally at VotingWorks, with 5 days including printing. It followed a test protocol aligned with VVSG requirement 2.7-C, but at a lower fidelity than in a Voting System Test Laboratory (VSTL) to accommodate 8-hour work days while still targeting highest risk failure modes. The system experienced no failures, but the HP4001dn printer saw some anomalies that require procedural mitigations and updated documentation.

Applicable VVSG Requirements

Primary VVSG requirement tested:

- 2.7-C: Continuous operation – varied environmental conditions

Supporting VVSG requirements tested:

- 1.1.6-F: Ability to clear misfed ballots
- 1.2-C: Minimum ballot positions
- 1.2-D: Handle maximum volume
- 1.2-E: Respond gracefully to stress of system limits
- 1.2-G: Misfeed rate benchmark
- 2.7-A: Assessment of reliability
- 2.7-B: Continuous operation - typical environmental conditions

Devices Under Test

- VxMark ballot marking device (BMD)
- HP4001dn laser printer
- Goldenmate 1500VA/1000W uninterruptible power supply (UPS)

Purpose

This test intended to stress the VxMark system with more extreme but realistic operational conditions and evaluate its performance. The HP4001dn printer was the main hardware component of the system to evaluate, because there was limited information to explain how well it performs in the full operational ambient temperature range of 50°F to 95°F required by VVSG.

The VxMark BMD and Goldenmate 1500VA/1000W UPS were not expected to show any issues. The VxMark BMD shares many components with VxScan and also has fewer temperature-sensitive components than VxScan, and VxScan previously passed similar continuous operational tests at VotingWorks and at Element in Longmont, CO. The Goldenmate 1500VA/1000W UPS is also rated for a wider operational temperature range than VVSG specifies.

Subgoals of testing were to:

1. Give confidence that VxMark can meet the VVSG requirement **2.7-C: Continuous operation – varied environmental conditions**.
 - a. The current reported test adapted the VVSG 2.7-C procedure to be more practical to do outside a VSTL. It took place across 8 days, with warm cycles in the daytime during the weekdays, and cool cycles at night and on the weekend for the printer specifically. Operational tests focused on the highest risk operations for the VxMark system: printing in the coldest and warmest conditions, and printing at the temperature transitions. These procedures draw out similar system vulnerabilities as the official VVSG 2.7-C procedure, which is a 104-hour (4-5 day) continuous test, consisting of 3 days of alternating 12-hour cool and 12-hour warm cycles (50°F/25%RH and 95°F/55%RH), plus about 2 days of room temperature cycles.
 - b. Due to limitations in equipment and space, cool cycles would not control humidity. Only the printer would experience both cool and warm cycles. The VxMark BMD would experience cycles between warm and room temperature conditions. The UPS would only experience room temperature conditions.
2. Give confidence that VxMark can meet **VVSG 1.2-E: Respond gracefully to stress of system limits**.
 - a. This test could show the system response over time, after being on and printing for several days without any power cycling.
 - b. It would analyze how well the laser printer functions in more extreme cold and warm temperatures, and how well the VxMark BMD performs in warm temperatures.
3. **Find failure modes for the system at higher use**, whether in hardware or software.
4. **Check the temperature of the system over time**. This test could confirm previous testing showing safe external touch temperatures of the system over time and in different conditions.

Materials

- HP4001dn laser printer (VNL0728678), with:
 - compatible 28lb and 32lb bond paper
 - HP148A toner
- VxMark unit (MK-14-018) for system testing
 - Imaged with v4.0.4 production app
 - Configured with summary ballot test election
- Goldenmate 1500VA/1000W UPS (JUO2HE390707)
- VxScan unit for verifying printed ballots
- USB stick with election definition
- Smart cards configured for election
- Warm climate-control space
 - Plastic Greenhouse Tent (8' x 6' x 7.5', Crown Shades)
 - Smart space heater (Dreo 1500W)
 - Smart thermometer-hygrometer (Govee H5075)
 - Smart humidifier (Govee 3L)
 - Govee control app
 - Table, high-amperage extension cable
- Cold climate-control space
 - Freezer (Insignia 14 cu. ft.)
 - Temperature controller thermostat (Inkbird WiFi ITC-308)
 - Plastic wrap, bubble wrap, tape, supporting material
- Thermal camera (Topdon TC004 Mini)

Procedures

Specific test steps were as follows:

1. Plan for cool and warm cyclic operations.

- a. Plan for the system to experience the following conditions while operational:

Test Days	Operational Conditions
1-3 (Friday, Saturday, Sunday)	● Printer: ○ Evening: 50°F ● VxMark BMD: ○ Evening: Room temperature ● UPS: ○ Evening: Room temperature

4-7 (Monday, Tuesday, Wednesday, Thursday)	● Printer: ○ Morning: 50°F, print ○ Daytime: 95°F, print every hour ○ Evening: 50°F ● VxMark BMD: ○ Morning: Room temperature ○ Daytime: 95°F, vote every hour ○ Evening: Room temperature ● UPS: ○ Room temperature
8 (Friday)	● Printer: ○ Morning: 50°F, print ○ Daytime: 95°F, print every hour ○ Evening: Shutdown ● VxMark BMD: ○ Morning: Room temperature ○ Daytime: 95°F, vote every hour ○ Evening: Shutdown ● UPS: ○ Room temperature ○ Evening: Shutdown

- b. Connect the system, with the VxMark BMD and printer powered by the UPS, which is connected to mains power.

2. Set up the cool operation chamber.

- a. When testing cool operation, store the printer in the chest freezer, with its power and data cables still connected outside the freezer. Control the temperature at 50°F via a thermostat with a temperature probe inside. Since the cables cause the freezer lid to stay slightly open, seal the opening with plastic wrap to minimize temperature loss.
- b. Note that ending a cooling cycle means that the freezer (and/or test sample inside the freezer) should go up to temperature across a period of at least 30 minutes, to better meet test standards and avoid unrealistic condensation.
- c. Example photos of the cool operation setup for the printer are shown below:





3. Set up the warm operation chamber.

- a. When testing warm operation, store the printer and VxMark BMD inside a climate tent with a smart heating element and humidifier. Leave the power and data cables connected properly to the UPS, which remains outside the tent. Control the temperature at 95°F via a thermostat with a temperature probe inside. Follow safety best practices to keep heating elements away from all objects, limited in output, powered only with cables rated for sufficiently high amperage, and regularly monitored.
- b. Note that ending a warm cycle means cooling the chamber (and/or test sample inside the chamber) to temperature across at least 30 minutes, to better meet test standards and avoid unrealistic failure modes.
- c. Example photo of warm operation setup is shown below (printer behind BMD):



4. Test and observe standard operations throughout each work day every hour.
 - a. Test printing at the start of the day while the printer is cool.
 - b. Test printing multiple batches of ballots throughout the daytime while the printer and VxMark BMD are warm.
 - c. Regularly monitor other VxScan software functions, including touchscreen functions, audio output, PAT input, ATI input, USB connections, card reader input, and barcode scanner or barcode reader function, as listed in *VxMark Tests of Normal Function*.
 - d. Probe the surface temperatures of the VxScan unit while in warm operation.
 - e. Investigate any anomalies or failures that arise and if mitigations are possible.

Results

1. The system operated for 8 consecutive days from 50°F to 95°F without power cycling.

- a. The printer experienced 5 cycles of cool 50°F and warm 95°F ambient operational conditions across 8 days.
 - i. Approximately 136 cumulative hours at 50°F.
 - ii. Approximately 32 cumulative hours at 95°F-55%RH.
 - iii. 408 total ballot prints were made, across 35 batches of prints.
 1. 15% were printed in cool conditions; 85% were printed in warm conditions.
 2. 87% were QR-code summary ballots; 13% were duplex prints resembling hand-marked paper ballots.
 3. 63% were printed on 32 lb bond paper weight; 37% were printed on 28 lb bond paper weight.
- b. The VxMark BMD experienced 5 cycles of room temperature and 95°F-55%RH ambient operational conditions across 8 days.
 - i. Approximately 136 cumulative hours at room temperature, or about 73°F.
 - ii. Approximately 32 cumulative hours at 95°F and 55%RH.
- c. The UPS experienced room temperature conditions across 8 days.
 - i. Approximately 168 cumulative hours at room temperature, or about 74°F.
- d. A log of chronological observations throughout the week is attached in [Appendix A](#).

2. The VxMark BMD saw normal operation throughout testing.

- a. The touchscreen, audio output, PAT input, ATI input, USB connections, card reader input, and barcode scanner or barcode reader all showed normal function throughout regular checks during testing. This was true both at room temperature and in the warm temperature cycles
- b. The maximum temperature on the external surface of the VxMark unit was measured as 131°F, during operation at 95°F ambient air temperature, on the surface below the ATI above the SBC. This is below the 140°F maximum safe touch temperature standard recommended by OSHA.

3. The UPS saw normal operation throughout testing.

- a. The UPS remained on and powered the VxMark BMD and printer throughout the 8 days of testing, with no loss in functionality during or after the test.
- b. Short warning beeps occasionally sounded from the UPS, which indicated momentary power output greater than 1000 W but less than 1100 W, according to

the manufacturer user manual. This occurred at the start of 8 out of 35 printing sessions (20% of printing sessions) or 8 out of 408 prints (2% of prints) during testing. It consistently occurred at the very first prints after the printer was soaked overnight in a 50°F cool cycle, and it also occurred twice during warm cycles.

Beeps stopped after the first prints in a batch, as the laser printer internal mechanisms warmed up and drew less power on subsequent prints.

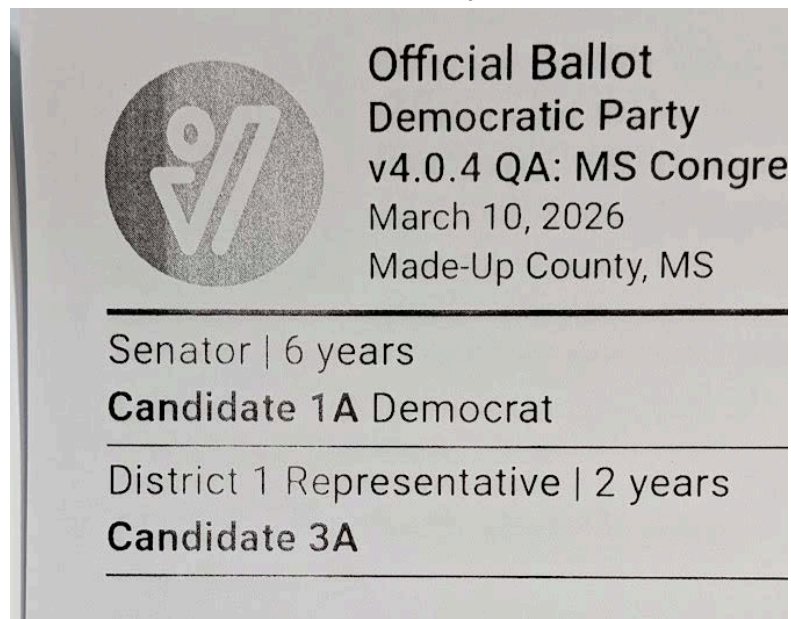
- c. When the VxMark system was idle, power draw from the UPS was very low, hovering close to 20W. This causes the numeric indicators of power output on the UPS display to flash on and off, as the power output intermittently dips below the threshold that the UPS uses to display it. This flashing was normal behavior for such low power use.

4. The HP4001dn printer saw three anomalies in cooler and warmer temperatures that needed procedural mitigation.

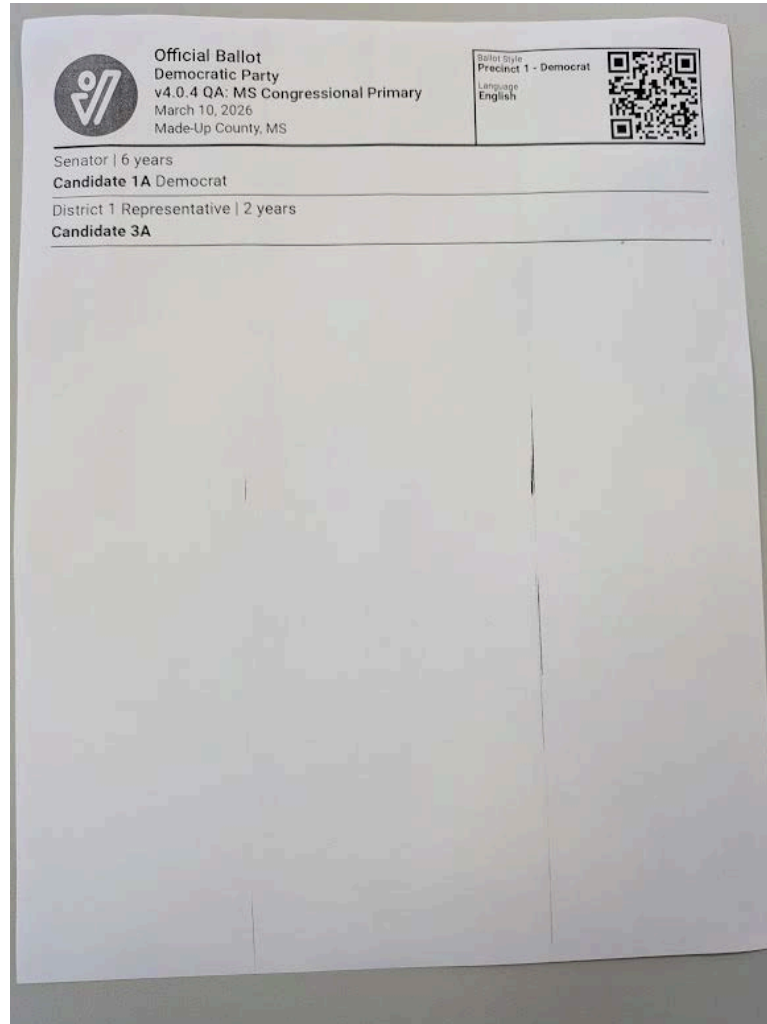
- a. The three anomalies in order of importance were (1) degraded print quality during 50°F operation or when transitioning from cool to warm operations, (2) paper not gripping after transitioning from cool to warm operations, and (3) slight layer of gray over the print area during 95°F operation. The first anomaly was the most major concern, because it consistently happened across multiple cool cycles and could potentially disrupt ballot scanning if not mitigated. The other two anomalies were smaller concerns. Details on each anomaly are given below.
- b. Anomaly 1: Degraded print quality during 50°F operation and transitioning from 50°F to 95°F. Ballots with this degraded print quality were all scanned and tallied properly in VxScan, but they had the potential to cause failures and did not go away when changing the toner. Several examples are described below.
 - i. After a transition from 50°F to 95°F faster than 30 minutes (against protocol), one ballot showed inked areas with slight distortion consistent with liquid or condensation being present upon printing. It scanned properly due to QR code robustness. Photos are shown here:



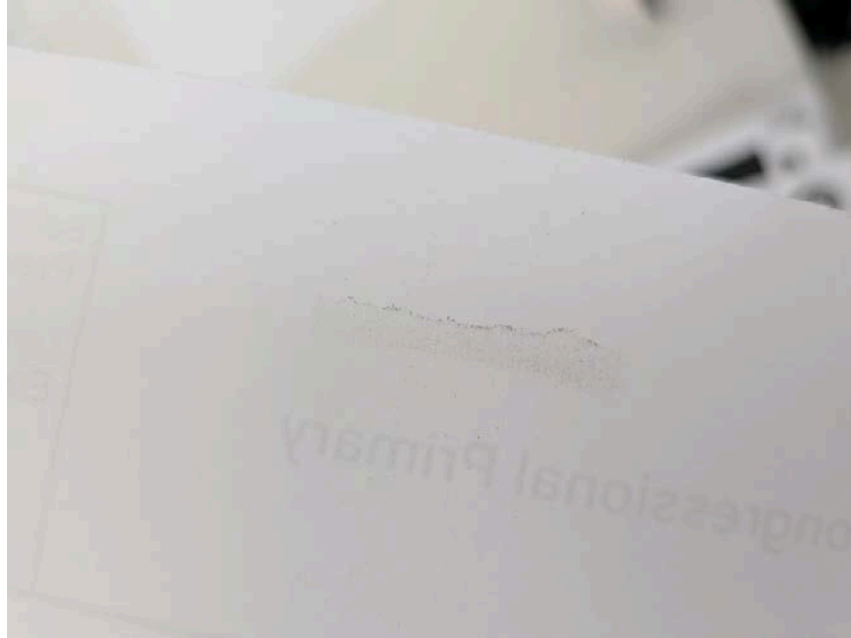
- ii. After a transition from 50°F to 95°F on multiple test days, several ballots showed some vertical areas of partly faded ink or toner, shown below:



- iii. After a transition from 50°F to 95°F on the last day of testing (Friday), one ballot showed some vertical streaking, consistent with loose ink or toner, or liquid mixing with ink inside the printer:



- c. Anomaly 2: Paper not gripping when transitioning from cool to warm conditions quickly.
 - i. The first print in this test was accidentally done against recommended protocol, transitioning the printer from cool to warm conditions instantly instead of over a standard 30-minute time span. When this happened, the printer failed to pick up the paper from the standard tray (Tray 2) and gave an error (event code 13.02.1). Inspection of the paper tray showed that condensation was present on the front pickup roller. The paper itself also had a wet spot on it where the pickup roller contacted it, reducing friction against the roller (shown below):



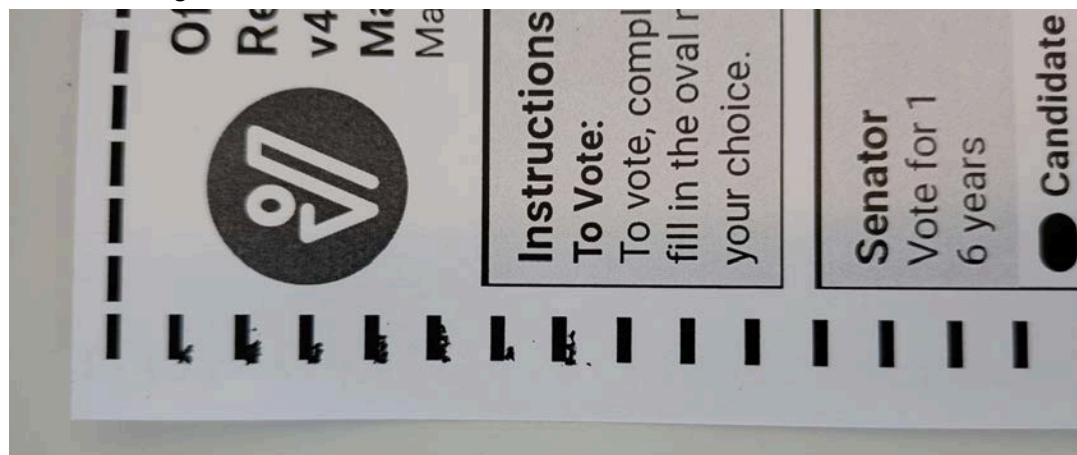
Wiping the condensation from the pickup roller and removing this old wet paper and replacing it with a fresh sheet allowed normal printing operation. Standard protocols were followed more closely following this anomaly, to transition between temperatures more slowly.

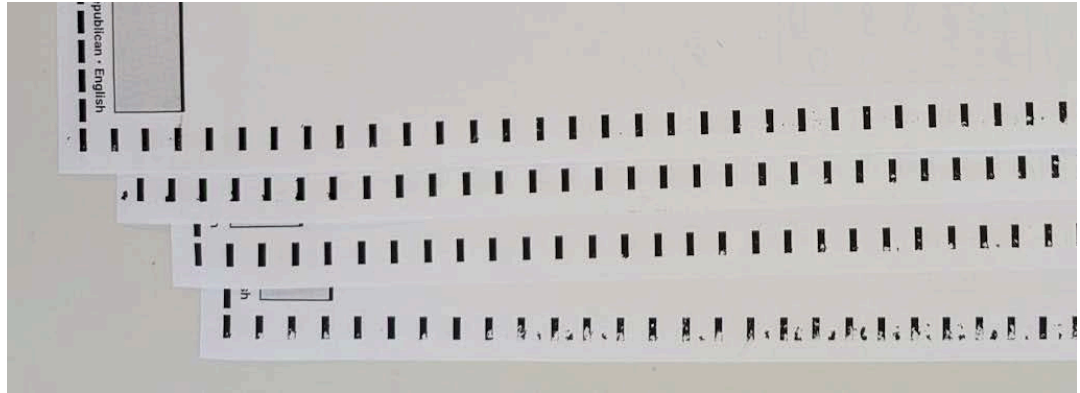
- d. Anomaly 3: Slight layer of gray over the print area during 95°F operation.
 - i. This appeared consistently after the printer was left for a few hours in the warm environment. It went away when transitioning back to cold temperatures. It did not affect scanning or tallying in VxScan. This is similar to a phenomenon previously seen after 4 hours of non-operational hot storage at 140°F (see *Test Report - VxMark Cold and Hot Storage of Goldenmate 1500VA/1000W UPS and HP4001dn Printer*); but in that case, the gray layer persisted even after cooling back to room temperature and had to be mitigated by replacing the HP148A toner. The layer of gray in this test was only temporary and did not require replacing toner. An example photo zoomed in on the light ink layer is shown below:



5. The HP4001dn required modified hardware settings when printing HMPBs in 50°F environments.

- a. Additional tests were performed after completing the initial 8 days of testing to further investigate the first printer anomaly. The printer was left in the 50°F overnight soak and configured to print more duplex full HMPBs in the cool environment, to see the effects on printing timing marks on the sides of the paper. This specific scenario was not previously investigated in the original 8 test days.
- b. Printing while cold consistently led to faulty HMPBs. Examples shown below have distorted timing marks that would be rejected by VxScan. The distortions could be either additional blotches of ink along the timing marks or missing ink in the real timing marks, as shown below:





- c. Several mitigation strategies did not work while the printer remained cold; these included using the internal printer cleaning function, changing paper, changing toner, and printing many sheets to warm up the internals. These distortions went away when bringing the printer back up to room temperature and letting it sit for at least two hours.
- d. A consistent mitigation that restored print quality while the printer was still cold was to change the hardware settings of the printer itself in its configuration menu, using the buttons on the front of the printer. The settings were changed to slow down printing, by choosing:

**Main Menu → Setup → Tray Management →
Default Paper Type → Mid-Weight 96-110g**

This consistently solved the issue. This was verified through additional cold soak testing of the printer and alternating between “Default Paper Type” settings. This mode of printing was about half as fast as the default setting. Note that other printer hardware menu setting changes did not help, and so other settings typically *should not* be changed.

Conclusions

- The VxMark system is likely to pass testing defined in VVSG 2.7-C, as long as procedural mitigations are properly followed for the printer.
- Documentation and training for VxMark should be updated to include the following:
 - When moving the HP4001dn printer from a cold storage environment (<60°F) to a warmer one (>60°F), let the printer sit at the warmer temperature for at least an hour before printing to avoid problems related to condensation. Do not immediately move the HP4001dn printer from a cold environment to a warm one and turn it on. If the printer must be suddenly moved, then wipe down the internal tray roller and replace the paper inside if it was also previously stored in the cold environment.
 - If operating the HP4001dn printer in an environment colder than 60°F, configure the printer hardware to print with better quality by adjusting this setting with the front menu:

■ **Main Menu → Setup → Tray Management → Default
Paper Type → Mid-Weight 96-110g**

This setting can be applied preemptively, or it can be set in response to seeing print quality issues similar to those pictured in this report.

- If print issues are still seen, other mitigation options are to print cleaning pages, replace paper, replace toner, and to wipe down or clean the tray rollers.

Appendix A: Summary Log of Observations

Time & Conditions				Cumulative hours in conditions						Ambient conditions					No. of ballot prints						
Date	Day	Time	Printer ambient	Cold, printer	Warm, printer	Warm, VxMark	Room temp, VxMark	Room temp, UPS	Test action	Temp (F), printer	%RH, printer	Temp (F), VxMark	%RH, VxMark	Temp (F), UPS	In batch	Cumulative	Cumulative, in cold	Cumulative, in warm	UPS behavior	thermal behavior	Notes
5/1/2026	Fri	5:00 PM	cold	0	0	0	0	0	setup cold	50	56	70		70		0	0	0			
5/4/2026	Mon	10:48 AM	cold to warm	65.8	0	0	65.8	65.8	print cold	54	67	76		76	13	13	13	0	Warning beep first print		
5/4/2026	Mon	10:54 AM	warm	65.90	0	0	65.90	65.90	setup warm							13	13	0			
5/4/2026	Mon	11:50 AM	warm	65.90	0.93	0.93	65.90	66.83	print warm	92.3	54.5	92.3	54.5	76	12	25	13	12			First print needed 3x try due to error "unable to pick paper from Tray 2, Event Code 13.03.01"
5/4/2026	Mon	1:30 PM	warm	65.90	2.60	2.60	65.90	68.50	print warm	93	50.6	93	50.6	76	16	41	13	28	no issues		
5/4/2026	Mon	2:20 PM	warm	65.90	3.43	3.43	65.90	69.33	print warm	94	53.4	94	43.4	76	12	53	13	40	no issues		prints starting to have slight grayness over whole page from toner
5/4/2026	Mon	3:04 PM	warm	65.90	4.17	4.17	65.90	70.07	print warm	94.7	54.1	94.7	54.1	76	12	65	13	52	no issues		still slight gray layer
5/4/2026	Mon	4:24 PM	warm	65.90	5.50	5.50	65.90	71.40	print warm	94.6	53	94.6	53	76	12	77	13	64	no issues	131F max external touch temp, metal under ATI; similar temps around front right area	had to load more paper
5/4/2026	Mon	4:37 PM	warm to cold	65.90	5.72	5.72	65.90	71.62	setup cold							77	13	64			
5/4/2026	Mon	5:30 PM	cold	66.78	5.72	5.72	66.78	72.50	monitor	72.7	18.7	76		76		77	13	64		Printer at 72.7F while ambient at 50F at 5:30pm	

Time & Conditions				Cumulative hours in conditions						Ambient conditions					No. of ballot prints						
Date	Day	Time	Printer ambient	Cold, printer	Warm, printer	Warm, VxMark	Room temp, VxMark	Room temp, UPS	Test action	Temp (F), printer	%RH, printer	Temp (F), VxMark	%RH, VxMark	Temp (F), UPS	In batch	Cumulative	Cumulative, in cold	Cumulative, in warm	UPS behavior	thermal behavior	Notes
5/5/2026	Tue	9:34 AM	cold	82.85	5.72	5.72	82.85	88.57	print cold	53.6	49.8	71		71	11	88	24	64	Warning beep first print		no gray layer on prints anymore in cold; 1st print had streaks maybe from condensation
5/5/2026	Tue	9:39 AM	cold to warm	82.85	5.80	5.80	82.85	88.65	setup warm							88	24	64			
5/5/2026	Tue	9:40 AM	cold to warm	82.85	5.82	5.82	82.85	88.67	monitor	72	64	72		72		88	24	64			
5/5/2026	Tue	10:34 AM	warm	82.85	6.72	6.72	82.85	89.57	print warm	87.6	53	87.6	53	72	10	98	24	74	no issues		
5/5/2026	Tue	11:37 AM	warm	82.85	7.77	7.77	82.85	90.62	print warm	93.9	53	93.9	53	74	15	113	24	89	no issues		
5/5/2026	Tue	12:20 PM	warm	82.85	8.48	8.48	82.85	91.33	print warm	94.9	53	94.9	53	75	12	125	24	101	no issues		
5/5/2026	Tue	1:20 PM	warm	82.85	9.48	9.48	82.85	92.33	print warm	94.7	54	94.7	54	75	12	137	24	113	no issues		
5/5/2026	Tue	2:20 PM	warm	82.85	10.48	10.48	82.85	93.33	print warm	95.5	53	95.5	53	76	13	150	24	126	no issues	120F max at 2:24pm under ATI	
5/5/2026	Tue	3:31 PM	warm	82.85	11.67	11.67	82.85	94.52	print warm	94.7	54	94.7	54	76	11	161	24	137	no issues	123F max at 3:31pm under ATI	
5/5/2026	Tue	4:31 PM	warm	82.85	12.67	12.67	82.85	95.52	print warm	94.5	54	94.5	54	76	14	175	24	151	no issues		
5/5/2026	Tue	4:36 PM	warm to cold	82.93	12.67	12.67	82.93	95.60	setup cold							175	24	151	Noticed low power draw when idle		started by opening tent, then placing in freezer.
5/6/2026	Wed	9:45 AM	cold	100.08	12.67	12.67	100.08	112.75	print cold	54.2	58.9	74		74	12	187	36	151	Warning beep first print, 1053W output		
5/6/2026	Wed	10:23 AM	cold to warm	100.08	13.30	13.30	100.08	113.38	setup warm	74.1	62.3	74.1	62.3	74		187	36	151			

Time & Conditions				Cumulative hours in conditions						Ambient conditions					No. of ballot prints						
Date	Day	Time	Printer ambient	Cold, printer	Warm, printer	Warm, VxMark	Room temp, VxMark	Room temp, UPS	Test action	Temp (F), printer	%RH, printer	Temp (F), VxMark	%RH, VxMark	Temp (F), UPS	In batch	Cumulative	Cumulative, in cold	Cumulative, in warm	UPS behavior	thermal behavior	Notes
5/6/2026	Wed	10:41 AM	warm	100.08	13.60	13.60	100.08	113.68	print warm	88.4	51	88.4	51	73	15	202	36	166	no issues		some print artifacts related to condensation, temperature changes
5/6/2026	Wed	11:37 AM	warm	100.08	14.53	14.53	100.08	114.62	print warm	93.1	54	93.1	54	73	13	215	36	179	no issues		no print issues
5/6/2026	Wed	12:51 PM	warm	100.08	15.77	15.77	100.08	115.85	print warm	93.9	53	93.9	53	74	12	227	36	191	no issues	127.4F max under ATI	no print issues; one VxScan scan failure from holding ballot too tightly causing skew
5/6/2026	Wed	1:31 PM	warm	100.08	16.43	16.43	100.08	116.52	print warm	94.2	53	94.2	53	75	12	239	36	203	no issues		
5/6/2026	Wed	3:10 PM	warm	100.08	18.08	18.08	100.08	118.17	print warm	93.8	53	93.8	53	75	12	251	36	215	no issues	127.3F max under ATI	light gray layer over prints
5/6/2026	Wed	4:00 PM	warm	100.08	18.92	18.92	100.08	119.00	print warm	93.5	54	93.5	54	75	2	253	36	217	Warning beep first print		light gray layer over prints
5/6/2026	Wed	4:49 PM	warm	100.08	19.73	19.73	100.08	119.82	print warm	80	65	80	65	75	4	257	36	221	Warning beep first print	115.1F max under ATI	light gray layer over prints
5/6/2026	Wed	4:52 PM	warm to cold	100.08	19.78	19.78	100.08	119.87	setup warm							257	36	221			loaded new paper: 28lb Office Depot, changed from 32lb HP Premium
5/7/2026	Thu	9:30 AM	cold	116.72	19.78	19.78	116.72	136.50	print cold	54.2	47.6	69		69	12	269	48	221	Warning beep first print		no issues printing, 28lb bond paper
5/7/2026	Thu	9:35 AM	cold to warm	116.72	19.87	19.87	116.72	136.58	setup warm							269	48	221			
5/7/2026	Thu	10:24 AM	warm	116.72	20.68	20.68	116.72	137.40	print warm	83.3	42	83.3	42	69	11	280	48	232	Warning beep first print		no issues printing
5/7/2026	Thu	11:15 AM	warm	116.72	21.53	21.53	116.72	138.25	print warm	94.6	53.2	94.6	53.2	70	9	289	48	241	no issues		1st print slight streaks through QR code
5/7/2026	Thu	12:34 PM	warm	116.72	22.85	22.85	116.72	139.57	print warm	95.5	53	95.5	53	71	12	301	48	253	no issues		no issues printing

[illegible]

[illegible]